SPARSH ROY

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RESEARCH EXPERIENCE

Massachusetts Institute of Technology — Research Affiliate *2025 – Present*

Across various labs during different time frames | Cambridge, MA

- Supports research operations across a distributed critical care data group of ~47 members, helping onboard new researchers and support projects from question formulation through analysis and write-up.
- Supports a multi-team study auditing evidence provenance in AI clinical decision-support systems: which sources they draw on, and which patient populations that evidence represents.
- Contributes to development work applying data science methods to healthcare use cases, including analysis tooling and evaluation workflows for clinical datasets.
- Contributes to Mantis, an AI-powered cognitive cartography and visual data-science platform, building backend extensions and tooling for agentic exploration of multimodal knowledge spaces.
- Contributes to MatrAIx, a population-scale world simulation running 8.3 billion persona agents, and to PersonaBench, a benchmark for evaluating persona agent behavior at scale; supported promotion for a release that reached viral distribution.
- Supports outreach and external communications, helping manage correspondence with collaborators, prospective researchers, and partner research groups.
- Contributing to a manuscript in preparation for submission to Humanities and Social Sciences Communications.

University of Illinois Urbana-Champaign — Research Assistant *2026 – Present*

Human–computer interaction; automated evaluation of research question novelty | Urbana, IL

- Supports technical development of the evaluation pipeline and analysis infrastructure for automatically assessing novelty in research questions.
- Contributing to a manuscript in preparation for iConference.

University of California, Santa Cruz — Researcher *2026 – Present*

Propagator networks and AI explainability | Santa Cruz, CA

- Works on propagator networks as a substrate for explainability, where propagation carries the provenance of a value alongside the value itself, so a system can account for how it reached an output rather than having an explanation reconstructed after the fact.
- Contributed for the problem's own sake rather than for authorship credit.

Google-Sponsored Research Laboratory — Researcher *2026 – Present*

Mathematics and machine-learning theory | Vienna, Austria

- Investigative and scholarly research across mathematics and machine learning.

SELECTED RESEARCH PROJECTS

When Do Learned Diffusion Proposals Help Constraint Solving? A Controlled Study on Continuous Algebraic Systems — arXiv:2607.27169; under review at AAAI.

- Compares graph-neural diffusion proposals against exact symbolic acceptance on continuous algebraic systems; reports a negative result under matched refinement budgets and a positive finding on learned repair ranking.

Belief-Space Perception Routing under Coupled Sensor Faults and Compute Contention — arXiv:2608.00322; under review at IEEE MIT URTC.

- Models perception routing under belief uncertainty where sensor faults and compute contention are coupled rather than independent.

Substrate-Coevolving Neural Dynamics: A Diagnostic Study of Self-Modifying Computation — first author; SSRN:6892259.

- Makes the computational interaction substrate an explicit state variable updated during the forward pass, then diagnoses the resulting failure modes, optimization barriers, and gated, latch-like substrate updates.

KAISER: Reproducible Subgroup Fairness Auditing for Clinical Risk Models — arXiv:2607.28608.

- Fixes an auditing protocol so subgroup performance gaps in clinical risk models can be rerun and compared across studies rather than re-derived ad hoc.

MOSAIC — The Garden of Forking Tissues: A Specification-Curve Multiverse for Spatial Transcriptomics.

- Sweeps a multiverse of defensible pipeline specifications and quantifies how much of a biological conclusion survives reasonable analytic choices.

Aura Farming the Good Life: Toward a Community-Based Epistemology of Learning in the Age of AI — open-science manuscript; in submission (Nature portfolio).

- Argues for community-based epistemology as an alternative to credential-driven academic culture.

AidSense — Crisis-Resource Access Platform

- AI-assisted platform connecting individuals to local health and social services, with offline access and interactive mapping. Congressional App Challenge winner; showcased at the U.S. Capitol.

AWARDS & RECOGNITION

- Bronze Medalist, United States Medical Disease Olympiad — top 150 nationally.
- Congressional App Challenge Winner, U.S. House of Representatives (2025) — AidSense; showcased at #HouseOfCode 2026.
- NJ INNOVATE100 Top Innovator (2026), sponsored by Rutgers, Princeton, and Nokia; among the youngest recipients selected.

EDUCATION

Hopewell Valley Central High School — High School Diploma *Expected Spring 2028*
Princeton, NJ | GPA: 4.25 weighted

Syracuse University — Dual Enrollment *2026 – Present*
Calculus III (MAT 397); mathematics coursework through multivariable calculus

MIT Beaver Works Summer Institute — Cognitive Assistants *Summer 2025*
Selective summer program in autonomous systems and applied AI

TECHNICAL SKILLS

Clinical & biomedical research: Retrospective cohort design; human-subjects protocols and IRB; evidence provenance auditing; systematic literature screening; clinical decision-support evaluation; computational oncology; spatial transcriptomics (scanpy, squidpy); single-cell analysis.

ML, HCI & statistics: Supervised and generative modeling; benchmark design and model evaluation; agent-based simulation; interpretability and explainability; human-subjects study design; statistical inference and multiplicity control; specification-curve and sensitivity analysis.

Languages & tools: Python (pandas, scikit-learn, PyTorch), R, MATLAB, JavaScript/TypeScript, SQL, Git, AWS.